

# ESC TEST MODE

Quick guide for dual BLDC ESC feature testing

NXP Cup Car  
Standalone test screen

**Safety first:** Start with wheels off the ground. Potentiometer speed is capped to 30 percent in this test.

## 1. Enter and Arm

- Enter:** Open the NXP Cup tests menu and select ESC.
- Initial state:** The test starts DISARMED and sends neutral to both ESCs.
- Arm:** Hold SW2. The car enters ARM for about 3 seconds while neutral pulses are sent.
- Ready:** When the screen shows RDY, return the potentiometer to zero before applying throttle.
- Run:** Increase the potentiometer slowly. The state changes to RUN only after the zero check is satisfied.

## Button Controls

- SW2 hold

Arm from DIS/STOP; disarm from RDY/RUN/ARM.
- SW2 click

Cycle output mode: BOTH, ESC1, ESC2, DIFF.
- SW3 hold

Emergency stop. Both ESCs return to neutral.
- SW3 click

Toggle direction: FWD or REV.
- Pot

Speed magnitude. Low values are deadbanded; max command is 30 percent.

## OLED Screen Meaning

ESC TEST RDY  
M:BOTH FWD  
E1: -18 E2: -18  
P:153 C: -18%

DIS/ARM/RDY/RUN/STOP

Current test state.

M

Selected output mode.

FWD/REV

Direction sign applied to pot command.

E1/E2

Actual commands sent to primary and secondary ESCs.

P/C

Raw pot level and capped command percent.

## Output Modes

| Mode | ESC1 command    | ESC2 command     | Use it for                                      |
|------|-----------------|------------------|-------------------------------------------------|
| BOTH | same as command | same as command  | Basic same-speed motor test.                    |
| ESC1 | command         | neutral          | Check primary ESC/motor alone.                  |
| ESC2 | neutral         | command          | Check secondary ESC/motor alone.                |
| DIFF | command         | opposite command | Differential steering / wiring direction check. |

## LED Indication

- **DIS**  
Disarmed

**ARM**  
Arming neutral pulses

**RDY**  
Armed, waiting at neutral

**RUN**  
Commanding ESC outputs

**STOP**  
Emergency stop / neutral